AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109

.Net Programming

Mid-term Examination – (MTA)
Fall Semester 2025/2026

Date: 18/Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 90 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1- What is the primary difference between StreamReader and StreamWriter?

a) StreamReader writes data, StreamWriter reads data
b) StreamReader reads text files, StreamWriter writes text files
c) StreamReader handles binary data only
d) StreamWriter cannot create new files

2- Which statement about CheckBox controls is TRUE?

a) Only one CheckBox can be selected at a time
b) CheckBoxes support independent selection
c) CheckBoxes must be placed in a GroupBox to function
d) CheckBoxes do not have a Checked property

3- What happens when you assign an int to a double without casting?

a) Compilation error occurs
b) Implicit conversion happens automatically
c) The value is lost
d) Runtime exception is thrown

4- What does the Parse() method do when given invalid input?

a) Returns null
b) Returns zero
c) Throws an exception
d) Returns the original string

5- Which property determines the items displayed in a ComboBox?

a) Items
b) DisplayItems
c) ListSource
d) DataItems

6- When debugging a C# application in Visual Studio, you can pause execution at a specific line using a _____.

a) Stop point
b) Breakpoint
c) Pause marker
d) Debug line

7- In Windows Forms, a control that allows users to select a date from a calendar is called a _____.

a) Calendar Control
b) DateSelector
c) DateTimePicker
d) TimeControl

8- What is the result of the following code snippet? double result = (double)15 / 2;

a) 7
b) 7.5
c) 8
d) None of the mentioned

9- A string data type can be converted to an int using implicit casting.

a) True b) False

10- A program with a syntax error will compile successfully but may crash during execution.

a) True b) False

Part 2: Short Questions [20 marks]

Q1- Explain the difference between File.ReadAllText() and File.ReadAllLines(). When would you use each?

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Q2- Given this if-else-if statement, rewrite it using a switch statement with equivalent functionality.

```
if (paymentMethod == "Credit")
{   MessageBox.Show("Processing credit card");  }
else if (paymentMethod == "Debit")
{   MessageBox.Show("Processing debit card");  }
else
{   MessageBox.Show("Invalid payment method");  }
```

Q3- What is the output of the following program? And what will be the output if we replace the shaded word with "break"?

```
int sum = 0;
for (int i = 1; i <= 6; i++)
{ if (i % 2 == 0) continue;
  sum += i;
  MessageBox.Show(System.Convert.ToString(sum));}
MessageBox.Show("Total: " + sum);
```

Q4- Complete the following paragraph with the suitable terms.

[runtime error, continue, validation, syntax error, debug, Name, double, if-else, variable, enabled, visible, readonly]

1- A(n) _____ prevents the program from compiling until it is fixed.
2- When you pause a program to inspect variables, you are in _____ mode.
3- The _____ property of a control determines its unique identifier in code.
4- To check if user input is valid before processing, you perform _____.

Q5- Trace the following code. Specify the result of each line not only the final output.

```
using System;
namespace CodeTracing
{ class Program
  {  static void Main(string[] args)
     {  int x = 20;
        int y = x / 3;
        double z = Math.Pow(2, 3);
        Console.WriteLine($"Initial: x={x}, y={y}, z={z}");
        for (int j = 1; j <= 3; j++)
        { if (j % 2 == 0)
          { x -= 3;
            y += j;          }
          else
          { x += 4;
            z = x / 4.0;       }
          Console.WriteLine($"Iteration {j}: x={x}, y={y}, z={z}");
     }
        try
        {  Console.WriteLine("\nFinal Calculation:");
           Console.WriteLine(x / (y - 6));          }
        catch (Exception ex)
        {     Console.WriteLine(ex.Message);    }
  } } }
```


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Part 3: Problem Solving Questions [30 marks]

Q1- You are tasked with creating a Windows Forms application that converts temperature between Celsius and Fahrenheit. [15 marks]

Requirements:

- The application should have a TextBox for entering the temperature value to convert.
- There should be two RadioButton controls:
  - "Celsius to Fahrenheit" (default selected)
  - "Fahrenheit to Celsius"
- A "Convert" button that performs the conversion based on the selected RadioButton and displays the result in a Label (e.g., "77.00 °F").
- The application must include:
  - Input validation using TryParse() to handle non-numeric input.
  - A "Clear" button that clears the TextBox, Label, and resets the RadioButton selection.

Formulas:

- Celsius to Fahrenheit: $F = (C \times 9/5) + 32$
- Fahrenheit to Celsius: $C = (F - 32) \times 5/9$

Q2- Write a C# program using Windows Forms that:

- Reads text from a file named data.txt.
- Counts the total number of odd numbers and even numbers in the text.
- Writes the results to a file named report.txt in the format:

  Odd: [count]
  Even: [count]

- Your program should include a Button labeled "Process File" to trigger the process.
- Ensure your program correctly handles file I/O and provides feedback to the user (e.g., a MessageBox indicating completion or errors). [15 Marks]

End of questions

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